First-Order Characterization of Convex Minimizers

An End-to-End Meta-Math Walkthrough

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Abstract

A supporting affine function supplies global sufficiency. Feasible line segments supply the converse even when the minimizer is on the boundary. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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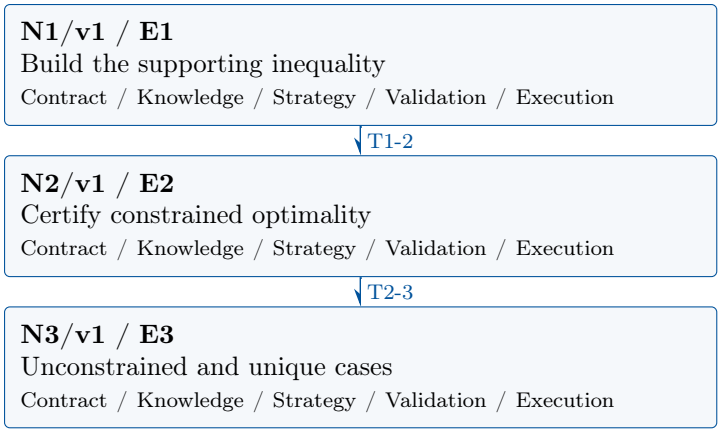
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, and let $C \subset \mathbb{R}^d$ be nonempty and convex. For $x_* \in C$, prove that x_* minimizes f over C if and only if $\langle \nabla f(x_*), y - x_* \rangle \geq 0$ for every $y \in C$. In particular, on $C = \mathbb{R}^d$ this is equivalent to $\nabla f(x_*) = 0$. If f is strictly convex on C, there is at most one minimizer.*

A. Foundations and permitted imports

Convexity means $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for $0 \leq t \leq 1$; strict convexity makes it strict for distinct points and $0 < t < 1$. Use differentiation along line segments and the elementary Euclidean inner product. This is a characterization at a proposed point, not an existence theorem for arbitrary C.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$.

Receiving activity. N2.2

Compatibility. E1/v1, convex C, and a proposed $x_* \in C$. Check: A boundary minimizer need not have zero gradient; only feasible directions are tested.

T2-3: N2/E2/v1 to N3. x_* is a global minimizer over C exactly when all feasible directional inner products are nonnegative.

Receiving activity. N3.1

Compatibility. E2/v1, and strict convexity when asserting uniqueness. Check: Uniqueness does not imply existence; for example a function on an open feasible set can have an unattained infimum.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Build the supporting inequality

Object and role. Divide the convex secant inequality by its positive parameter and let it tend to zero. **Local obligation:** $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$.

Inputs. Differentiable convex f and any $x, y \in \mathbb{R}^d$.

C — Contract

Publish E1/v1 only after establishing this obligation: $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$.

K — Knowledge

Differentiable convex f and any $x, y \in \mathbb{R}^d$. Retain the stated assumptions and import versions.

S — Strategy

Divide the convex secant inequality by its positive parameter and let it tend to zero.

V — Validation

The parameter approaches zero through positive values; a one-sided segment is sufficient.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 For $0 < t \leq 1$, convexity gives $f(x + t(y - x)) - f(x) \leq t(f(y) - f(x))$. Divide by $t > 0$.

N1.2 Differentiability at x makes the left quotient tend to $\langle \nabla f(x), y - x \rangle$ as $t \downarrow 0$. Rearrangement yields the supporting inequality for every x, y.

Export E1/v1. $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$.

Acceptance gate. The parameter approaches zero through positive values; a one-sided segment is sufficient.

Targeted repair. If a local Taylor approximation is used as a global lower bound, replace it by this convexity argument; Taylor's remainder has no automatic sign. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Certify constrained optimality

Object and role. Use feasible directional derivatives for necessity and the global support for sufficiency. **Local obligation:** x_* is a global minimizer over C exactly when all feasible directional inner products are nonnegative.

Inputs. E1/v1, convex C, and a proposed $x_* \in C$. Incoming proof exports: E1/v1 at N2.2.

C — Contract

Publish E2/v1 only after establishing this obligation: x_* is a global minimizer over C exactly when all feasible directional inner products are nonnegative.

K — Knowledge

E1/v1, convex C, and a proposed $x_* \in C$. Incoming proof exports: E1/v1 at N2.2. Retain the stated assumptions and import versions.

S — Strategy

Use feasible directional derivatives for necessity and the global support for sufficiency.

V — Validation

A boundary minimizer need not have zero gradient; only feasible directions are tested.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 If x_* minimizes f, then for $y \in C$ the segment $x_* + t(y - x_*)$ remains feasible. Its difference quotient from x_* is nonnegative for $t > 0$. Taking the limit gives $\langle \nabla f(x_*), y - x_* \rangle \geq 0$.

N2.2 Conversely, if those inequalities hold, E1 gives $f(y) \geq f(x_*) + \langle \nabla f(x_*), y - x_* \rangle \geq f(x_*)$ for every $y \in C$. This is the required global conclusion.

Export E2/v1. x_* is a global minimizer over C exactly when all feasible directional inner products are nonnegative.

Acceptance gate. A boundary minimizer need not have zero gradient; only feasible directions are tested.

Targeted repair. If stationarity is imposed at the boundary, restore the feasible-direction inequality instead of demanding all directions have zero derivative. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Unconstrained and unique cases

Object and role. Test the negative gradient as a direction, and use a midpoint to rule out two distinct minimizers. **Local obligation:** For $C = \mathbb{R}^d$, zero gradient is equivalent to optimality; strict convexity gives at most one minimizer.

Inputs. E2/v1, and strict convexity when asserting uniqueness. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: For $C = \mathbb{R}^d$, zero gradient is equivalent to optimality; strict convexity gives at most one minimizer.

K — Knowledge

E2/v1, and strict convexity when asserting uniqueness. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Test the negative gradient as a direction, and use a midpoint to rule out two distinct minimizers.

V — Validation

Uniqueness does not imply existence; for example a function on an open feasible set can have an unattained infimum.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 In the unconstrained case substitute $y = x_* - \nabla f(x_*)$ in E2 to get $-\|\nabla f(x_*)\|^2 \geq 0$, hence the gradient is zero. The converse follows immediately from E2.

N3.2 If distinct x_*, z_* both minimize and f is strictly convex on C, their midpoint lies in C and has strictly smaller value than the average of their equal minimum values, a contradiction. This proves at-most-one, completing the theorem.

Export E3/v1. For $C = \mathbb{R}^d$, zero gradient is equivalent to optimality; strict convexity gives at most one minimizer.

Acceptance gate. Uniqueness does not imply existence; for example a function on an open feasible set can have an unattained infimum.

Targeted repair. If a unique minimizer is declared without an existence argument, retain the precise at-most-one conclusion. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Build the supporting inequality

1. **N1.1.** For $0 < t \leq 1$, convexity gives $f(x + t(y - x)) - f(x) \leq t(f(y) - f(x))$. Divide by $t > 0$.

2. **N1.2.** Differentiability at x makes the left quotient tend to $\langle \nabla f(x), y - x \rangle$ as $t \downarrow 0$. Rearrangement yields the supporting inequality for every x, y.

B. N2 — Certify constrained optimality

1. **N2.1.** If x_* minimizes f, then for $y \in C$ the segment $x_* + t(y - x_*)$ remains feasible. Its difference quotient from x_* is nonnegative for $t > 0$. Taking the limit gives $\langle \nabla f(x_*), y - x_* \rangle \geq 0$.

2. **N2.2.** Conversely, if those inequalities hold, E1 gives $f(y) \geq f(x_*) + \langle \nabla f(x_*), y - x_* \rangle \geq f(x_*)$ for every $y \in C$. This is the required global conclusion.

C. N3 — Unconstrained and unique cases

1. **N3.1.** In the unconstrained case substitute $y = x_* - \nabla f(x_*)$ in E2 to get $-\|\nabla f(x_*)\|^2 \geq 0$, hence the gradient is zero. The converse follows immediately from E2.

2. **N3.2.** If distinct x_*, z_* both minimize and f is strictly convex on C, their midpoint lies in C and has strictly smaller value than the average of their equal minimum values, a contradiction. This proves at-most-one, completing the theorem.

□

V. WORKED ORCHESTRATION

For $f(x) = (x + 1)^2$ and $C = [0, \infty)$, choose $x_* = 0$. The derivative is 2, not zero, but $f'(0)(y - 0) = 2y \geq 0$ for all feasible y. The theorem certifies the boundary minimizer; strict convexity makes it unique.

A. Boundary and hypothesis audit

Without convexity of f, zero gradient need not indicate a minimum: $f(x) = -x^2$ has $f'(0) = 0$ and a strict maximum at zero. Existence also fails for the strictly convex $f(x) = x^2$ on $C = (0, 1)$, whose infimum is not attained.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 claim: a constrained optimum must satisfy $\nabla f = 0$. The worked boundary example refutes it. **Diagnosis:** Strategy tested infeasible negative directions. **Repair:** test only $y - x_*$ with $y \in C$ and prove the variational inequality. This replaces the invalid candidate with the exact §1 statement; the unconstrained zero-gradient corollary still holds separately.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Add normal cones or convex inequality constraints to develop constrained optimality without confusing boundary points with unconstrained stationary points.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Boyd and Vandenberghe, Convex Optimization for convexity, optimality, and descent; University of Maryland, Scientific Computing I for the numerical-analysis setting. The estimates used here are derived explicitly in §4.